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REPORT

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SoilTemp: A global database of near-surface temperature

Jonas J. Lembrechts[1] | Juha Aalto[2,3] | Michael B. Ashcroft[4,5] | Pieter De Frenne[6] | Martin Kopecký[7,8] | Jonathan Lenoir[9] | Miska Luoto[3] | Ilya M. D. Maclean[10] | Olivier Roupsard[11,12] | Eduardo Fuentes-Lillo[13,14,15] | Rafael A. García[13,14] | Loïc Pellissier[16,17] | Camille Pitteloud[16,17] | Juha M. Alatalo[18,19] | Stuart W. Smith[20,21] | Robert G. Björk[22,23] | Lena Muffler[24,25] | Amanda Ratier Backes[26,27] | Simone Cesarz[27,28] | Felix Gottschall[27,28] | Joseph Okello[29,30] | Josef Urban[31,32] | Roman Plichta[31] | Martin Svátek[31] | Shyam S. Phartyal[33,34] | Sonja Wipf[35,36] | Nico Eisenhauer[27,28] | Mihai Pușcaș[37] | Pavel D. Turtureanu[38] | Andrej Varlagin[39] | Romina D. Dimarco[40] | Alistair S. Jump[41] | Krystal Randall[42] | Ellen Dorrepaal[43] | Keith Larson[43] | Josefine Walz[43] | Luca Vitale[44] | Miroslav Svoboda[8] | Rebecca Finger Higgens[45] | Aud H. Halbritter[46] | Salvatore R. Curasi[47] | Ian Klupar[47] | Austin Koontz[48] | William D. Pearse[48,49] | Elizabeth Simpson[48] | Michael Stemkovski[48] | Bente Jessen Graae[20] | Mia Vedel Sørensen[20] | Toke T. Høye[50] | M. Rosa Fernández Calzado[51] | Juan Lorite[51] | Michele Carbognani[52] | Marcello Tomaselli[52] | T'ai G. W. Forte[52] | Alessandro Petraglia[52] | Stef Haesen[53] | Ben Somers[53] | Koenraad Van Meerbeek[53] | Mats P. Björkman[22,23] | Kristoffer Hylander[54] | Sonia Merinero[54] | Mana Gharun[55] | Nina Buchmann[55] | Jiri Dolezal[7,56] | Radim Matula[8] | Andrew D. Thomas[57] | Joseph J. Bailey[58] | Dany Ghosn[59] | George Kazakis[59] | Miguel A. de Pablo[60] | Julia Kemppinen[3] | Pekka Niittynen[3] | Lisa Rew[61] | Tim Seipel[61] | Christian Larson[61] | James D. M. Speed[62] | Jonas Ardö[63] | Nicoletta Cannone[64] | Mauro Guglielmin[65] | Francesco Malfasi[65] | Maaike Y. Bader[66] | Rafaella Canessa[66] | Angela Stanisci[67] | Juergen Kreyling[24] | Jonas Schmeddes[24] | Laurenz Teuber[24] | Valeria Aschero[68,69] | Marek Čiliak[70] | František Máliš[71] | Pallieter De Smedt[6] | Sanne Govaert[6] | Camille Meeussen[6] | Pieter Vangansbeke[6] | Khatuna Gigauri[72] | Andrea Lamprecht[73] | Harald Pauli[73] | Klaus Steinbauer[73] | Manuela Winkler[73] | Masahito Ueyama[74] | Martin A. Nuñez[75] | Tudor-Mihai Ursu[76] | Sylvia Haider[26,27] | Ronja E. M. Wedegärtner[20] | Marko Smiljanic[77] | Mario Trouillier[77] | Martin Wilmking[77] | Jan Altman[7] | Josef Brůna[7] | Lucia Hederová[7] | Martin Macek[7] | Matěj Man[7] | Jan Wild[7] | Pascal Vittoz[78] | Meelis Pärtel[79] | Peter Barančok[80] | Róbert Kanka[80] | Jozef Kollár[80] | Andrej Palaj[80] | Agustina Barros[69] | Ana C. Mazzolari[69] | Marijn Bauters[29] | Pascal Boeckx[29] | José-Luis Benito Alonso[81] |

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Shengwei Zong[82] | Valter Di Cecco[83] | Zuzana Sitková[84] | Katja Tielbörger[85] | Liesbeth van den Brink[85] | Robert Weigel[25] | Jürgen Homeier[25] | C. Johan Dahlberg[54,86] | Sergiy Medinets[87] | Volodymyr Medinets[87] | Hans J. De Boeck[1] | Miguel Portillo-Estrada[1] | Lore T. Verryckt[1] | Ann Milbau[88] | Gergana N. Daskalova[89] | Haydn J. D. Thomas[89] | Isla H. Myers-Smith[89] | Benjamin Blonder[90,91] | Jörg G. Stephan[92] | Patrice Descombes[16,17,93] | Florian Zellweger[93] | Esther R. Frei[35,93] | Bernard Heinesch[94] | Christopher Andrews[95] | Jan Dick[95] | Lukas Siebicke[96] | Adrian Rocha[97] | Rebecca A. Senior[98] | Christian Rixen[35] | Juan J. Jimenez[99] | Julia Boike[100,101] | Aníbal Pauchard[13,14] | Thomas Scholten[102] | Brett Scheffers[103] | David Klinges[104] | Edmund W. Basham[104] | Jian Zhang[105] | Zhaochen Zhang[105] | Charly Géron[1,106] | Fatih Fazlioglu[107] | Onur Candan[107] | Jhonatan Sallo Bravo[108] | Filip Hrbacek[109] | Kamil Laska[109] | Edoardo Cremonese[110] | Peter Haase[111,112] | Fernando E. Moyano[96] | Christian Rossi[36,113,114] | Ivan Nijs[1]

[1]Research Group PLECO (Plants and Ecosystems), University of Antwerp, Wilrijk, Belgium
[2]Finnish Meteorological Institute, Helsinki, Finland
[3]Department of Geosciences and Geography, University of Helsinki, Helsinki, Finland
[4]Centre for Sustainable Ecosystem Solutions, School of Biological Sciences, University of Wollongong, Wollongong, NSW, Australia
[5]Australian Museum, Sydney, NSW, Australia
[6]Forest & Nature Lab, Department of Environment, Ghent University, Melle-Gontrode, Belgium
[7]Institute of Botany of the Czech Academy of Sciences, Průhonice, Czech Republic
[8]Faculty of Forestry and Wood Sciences, Czech University of Life Sciences Prague, Prague 6 - Suchdol, Czech Republic
[9]UR 'Ecologie et Dynamique des Systèmes Anthropisées' (EDYSAN, UMR 7058 CNRS-UPJV), Univ. de Picardie Jules Verne, Amiens, France
[10]Environment and Sustainability Institute, University of Exeter, Penryn, UK
[11]CIRAD, UMR Eco&Sols, Dakar, Senegal
[12]Eco&Sols, Univ Montpellier, CIRAD, INRAE, IRD, Institut Agro, Montpellier, France
[13]Laboratorio de Invasiones Biológicas (LIB), Facultad de Ciencias Forestales, Universidad de Concepción, Concepción, Chile
[14]Instituto de Ecología y Biodiversidad (IEB), Santiago, Chile
[15]School of Education and Social Sciences, Adventist University of Chile, Chile
[16]Landscape Ecology, Institute of Terrestrial Ecosystems, Department of Environmental Systems Science, ETH Zürich, Zürich, Switzerland
[17]Unit of Land Change Science, Swiss Federal Research Institute WSL, Birmensdorf, Switzerland
[18]Department of Biological and Environmental Sciences, Qatar University, Doha, Qatar
[19]Environmental Science Center, Qatar University, Doha, Qatar
[20]Department of Biology, Norwegian University of Science and Technology, Trondheim, Norway
[21]Asian School of Environment, Nanyang Technological University, Singapore, Singapore
[22]Department of Earth Sciences, University of Gothenburg, Gothenburg, Sweden
[23]Gothenburg Global Biodiversity Centre, Gothenburg, Sweden
[24]Experimental Plant Ecology, Institute of Botany and Landscape Ecology, University of Greifswald, Greifswald, Germany
[25]Plant Ecology, Albrecht-von-Haller-Institute for Plant Sciences, University of Goettingen, Goettingen, Germany
[26]Institute of Biology/Geobotany and Botanical Garden, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany
[27]German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany
[28]Institute of Biology, Leipzig University, Leipzig, Germany
[29]Isotope Bioscience Laboratory – ISOFYS, Ghent University, Gent, Belgium
[30]Mountains of the Moon University, Fort Portal, Uganda
[31]Department of Forest Botany, Dendrology and Geobiocoenology, Mendel University, Brno, Czech Republic
[32]Siberian Federal University, Krasnoyarsk, Russia

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[33]School of Ecology and Environment Studies, Nalanda University, Rajgir, India
[34]Department of Forestry and NR, H.N.B. Garhwal University, Srinagar-Garhwal, India
[35]WSL Institute for Snow and Avalanche Research SLF, Davos, Switzerland
[36]Swiss National Park, Chastè Planta-Wildenberg, Zernez, Switzerland
[37]A. Borza Botanical Garden and Department of Taxonomy and Ecology, Faculty of Biology and Geology, Babeș-Bolyai University, Cluj-Napoca, Romania
[38]A. Borza Botanical Garden, Babes-Bolyai University, Cluj-Napoca, Romania
[39]A.N. Severtsov Institute of Ecology and Evolution, Russian Academy of Sciences, Moscow, Russia
[40]Grupo de Ecología de Poblaciones de Insectos, IFAB (INTA - CONICET), Bariloche, Argentina
[41]Biological and Environmental Sciences, Faculty of Natural Sciences, University of Stirling, Stirling, UK
[42]Centre for Sustainable Ecosystem Solutions, School of Earth, Atmospheric and Life Sciences, University of Wollongong, Wollongong, NSW, Australia
[43]Climate Impacts Research Centre, Department of Ecology and Environmental Sciences, Umeå University, Abisko, Sweden
[44]CNR - Institute for Mediterranean Agricultural and Forest Systems, Ercolano (Napoli), Italy
[45]Dartmouth College, Hanover, NH, USA
[46]Department of Biological Sciences and Bjerknes Centre for Climate Research, University of Bergen, Bergen, Norway
[47]Department of Biological Sciences, University of Notre Dame, Notre Dame, IN, USA
[48]Department of Biology and Ecology Center, Utah State University, Logan, UT, USA
[49]Department of Life Sciences, Imperial College London, Ascot, UK
[50]Department of Bioscience and Arctic Research Centre, Rønde, Denmark
[51]Department of Botany, University of Granada, Granada, Spain
[52]Department of Chemistry, Life Sciences and Environmental Sustainability, University of Parma, Parma, Italy
[53]Department of Earth and Environmental Sciences, Leuven, Belgium
[54]Department of Ecology, Environment and Plant Sciences and Bolin Centre for Climate Research, Stockholm University, Stockholm, Sweden
[55]Department of Environmental Systems Science, ETH Zurich, Zurich, Switzerland
[56]Faculty of Science, Department of Botany, University of South Bohemia, České Budějovice, Czech Republic
[57]Department of Geography and Earth Sciences, Aberystwyth University, Wales, UK
[58]Department of Geography, York St John University, York, UK
[59]Department of Geo-information in Environmental Management, Mediterranean Agronomic Institute of Chania, Chania, Greece
[60]Department of Geology, Geography and Environment, University of Alcalá, Madrid, Spain
[61]Department of Land Resources and Environmental Sciences, Montana State University, Bozeman, MT, USA
[62]Department of Natural History, NTNU University Museum, Norwegian University of Science and Technology, Trondheim, Norway
[63]Department of Physical Geography and Ecosystem Science, Lund University, Lund, Sweden
[64]Department of Science and High Technology, Insubria University, Como, Italy
[65]Department of Theoretical and Applied Sciences, Insubria University, Varese, Italy
[66]Ecological Plant Geography, Faculty of Geography, University of Marburg, Marburg, Germany
[67]EnvixLab, Dipartimento di Bioscienze e Territorio, Università degli Studi del Molise, Termoli, Italy
[68]Facultad de Ciencias Exactas y Naturales, Universidad Nacional de Cuyo, Cuyo, Argentina
[69]Instituto Argentino de Nivologiá, Glaciologiá y Ciencias Ambientales (IANIGLA), CONICET, CCT-Mendoza, Mendoza, Argentina
[70]Faculty of Ecology and Environmental Sciences, Technical University in Zvolen, Zvolen, Slovakia
[71]Faculty of Forestry, Technical University in Zvolen, Zvolen, Slovakia
[72]Georgian Institute of Public Affairs, Tbilisi, Georgia
[73]GLORIA Coordination, Institute for Interdisciplinary Mountain Research, Austrian Academy of Sciences (ÖAW) & Department of Integrative Biology and Biodiversity Research, University of Natural Resources and Life Sciences Vienna (BOKU), Vienna, Austria
[74]Graduate School of Life and Environmental Sciences, Osaka Prefecture University, Osaka, Japan
[75]Grupo de Ecología de Invasiones, INIBIOMA, CONICET/Universidad Nacional del Comahue, Bariloche, Argentina
[76]Institute of Biological Research Cluj-Napoca, National Institute of Research and Development for Biological Sciences, Bucharest, Romania
[77]Institute of Botany and Landscape Ecology, University Greifswald, Greifswald, Germany
[78]Institute of Earth Surface Dynamics, Faculty of Geosciences and Environment, University of Lausanne, Lausanne, Switzerland
[79]Institute of Ecology and Earth Sciences, University of Tartu, Tartu, Estonia
[80]Institute of Landscape Ecology, Slovak Academy of Sciences, Bratislava, Slovakia
[81]Jolube Consultor Botánico, Jaca, Spain
[82]Key Laboratory of Geographical Processes and Ecological Security in Changbai Mountains, Ministry of Education, School of Geographical Sciences, Northeast Normal University, Changchun, China
[83]Majella Seed Bank, Majella National Park, Lama dei Peligni, Italy

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[84]National Forest Centre, Forest Research Institute Zvolen, Zvolen, Slovakia
[85]Plant Ecology Group, Department of Evolution and Ecology, University of Tübingen, Tübingen, Germany
[86]County Administrative Board of Västra Götaland, Gothenburg, Sweden
[87]Regional Centre for Integrated Environmental Monitoring, Odesa National I.I. Mechnikov University, Odesa, Ukraine
[88]Research Institute for Nature and Forest (INBO), Brussels, Belgium
[89]School of GeoSciences, University of Edinburgh, Edinburgh, UK
[90]School of Life Sciences, Arizona State University, Tempe, AZ, USA
[91]Department of Environmental Science, Policy, and Management, University of California, Berkeley, Berkeley, CA, USA
[92]Swedish Species Information Centre, Swedish University of Agricultural Sciences, Uppsala, Sweden
[93]Swiss Federal Research Institute WSL, Birmensdorf, Switzerland
[94]TERRA Teaching and Research Center, Faculty of Gembloux Agro-Bio Tech, University of Liege, Gembloux, Belgium
[95]UK Centre for Ecology & Hydrology, Midlothian, UK
[96]Bioclimatology, University of Goettingen, Göttingen, Germany
[97]Department of Biological Sciences and the Environmental Change Initiative, University of Notre Dame, Notre Dame, IN, USA
[98]Woodrow Wilson School of Public and International Affairs, Princeton University, Princeton, NJ, USA
[99]ARAID Research and Development, Zaragoza, Spain
[100]Alfred Wegener Institute Helmholtz Center for Polar and Marine Research, Potsdam, Germany
[101]Geography Department, Humboldt-Universität zu Berlin, Berlin, Germany
[102]Chair of Soil Science and Geomorphology, Department of Geosciences, University of Tuebingen, Tuebingen, Germany
[103]Department of Wildlife Ecology and Conservation, University of Florida, Gainesville, FL, USA
[104]School of Natural Resources and Environment, University of Florida, Gainesville, FL, USA
[105]Zhejiang Tiantong Forest Ecosystem National Observation and Research Station, School of Ecological and Environmental Sciences, East China Normal University, Shanghai, China
[106]Biodiversity and Landscape, TERRA Research Centre, University of Liège, Gembloux Agro-Bio Tech, Gembloux, Belgium
[107]Faculty of Arts and Sciences, Department of Molecular Biology and Genetics, Ordu University, Ordu, Turkey
[108]Universidad Nacional de San Antonio Abad del Cusco, Cusco, Peru
[109]Department of Geography, Masaryk University, Brno, Czech Republic
[110]Climate Change Unit, Environmental Protection Agency of Aosta Valley, Aosta, Italy
[111]Senckenberg Research Institute and Natural History Museum Frankfurt, Gelnhausen, Germany
[112]Faculty of Biology, University of Duisburg-Essen, Essen, Germany
[113]Remote Sensing Laboratories, Department of Geography, University of Zurich, Zurich, Switzerland
[114]Research Unit Community Ecology, Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Birmensdorf, Switzerland

Correspondence
Jonas J. Lembrechts, Research Group PLECO (Plants and Ecosystems), University of Antwerp, 2610 Wilrijk, Belgium.
Email: jonas.lembrechts@uantwerpen.be

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Abstract

Current analyses and predictions of spatially explicit patterns and processes in ecology most often rely on climate data interpolated from standardized weather stations. This interpolated climate data represents long-term average thermal conditions at coarse spatial resolutions only. Hence, many climate-forcing factors that operate at fine spatiotemporal resolutions are overlooked. This is particularly important in relation to effects of observation height (e.g. vegetation, snow and soil characteristics) and in habitats varying in their exposure to radiation, moisture and wind (e.g. topography, radiative forcing or cold-air pooling). Since organisms living close to the ground relate more strongly to these microclimatic conditions than to free-air temperatures, microclimatic ground and near-surface data are needed to provide realistic forecasts of the fate of such organisms under anthropogenic climate change, as well as of the functioning of the ecosystems they live in. To fill this critical gap, we highlight a call for temperature time series submissions to SoilTemp, a geospatial database initiative compiling soil and near-surface temperature data from all over the world. Currently, this database contains time series from 7,538 temperature sensors from 51 countries

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across all key biomes. The database will pave the way toward an improved global understanding of microclimate and bridge the gap between the available climate data and the climate at fine spatiotemporal resolutions relevant to most organisms and ecosystem processes.

KEYWORDS

climate change, database, ecosystem processes, microclimate, soil climate, species distributions, temperature, topoclimate

1 | INTRODUCTION

Current ecological research increasingly deals with large-scale patterns and processes, with global databases of species distributions and traits becoming increasingly available (Bruelheide et al., 2018; Kattge et al., 2019; Kissling et al., 2018). Analyses of these patterns and processes—and their predictions under anthropogenic climate change—often rely on global climatic grids at coarse spatial resolutions interpolated from standardized weather stations that represent long-term average atmospheric conditions (Lembrechts, Nijs, & Lenoir, 2019). Moreover, sensors in these weather stations are shielded from direct solar radiation and located at ~2 m above a frequently mown lawn (free-air temperature or "macroclimate," Jarraud, 2008). These climatic grids thus ignore many climate-forcing processes that operate near the ground surface, at fine spatiotemporal resolutions, and in environments that vary in their exposure to winds, radiation and moisture ("microclimate," Bramer et al., 2018; Daly, 2006; Körner & Hiltbrunner, 2018). Importantly, while these microclimatic processes often operate at fine spatiotemporal resolutions, they can affect ecological relations both at the local and the global scales (Ashcroft, Cavanagh, Eldridge, & Gollan, 2014; De Frenne et al., 2013; Lembrechts, Lenoir et al., 2019). For example,

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they can potentially protect ground-dwelling biota against long-term climate variability, providing microrefugia for these species to survive in locations deemed unsuitable in models using climate data at coarse spatial resolutions, or buffer organisms against short-term extreme events (Bramer et al., 2018; De Frenne et al., 2013; Lenoir, Hattab, & Pierre, 2017; Suggitt et al., 2018). Microclimates can however also expose organisms to more extreme temperatures, in which case distribution models that ignore such microclimates may erroneously predict species survival instead of extinction (Pincebourde & Casas, 2019). To provide realistic forecasts of species distributions and performance, as well as of the functioning of the ecosystems they operate in, climate data that incorporate microclimatic processes, ideally measured in situ, are thus urgently needed (Körner & Hiltbrunner, 2018).

2 | HORIZONTAL AND VERTICAL FEATURES DRIVING MICROCLIMATE

The offset between micro- and macroclimate is particularly pronounced around the soil surface, as temperatures measured at 2 m above the ground can differ substantially from those at ground level, or in the layers just above and below it (Geiger, 1950; Lembrechts, Lenoir et al., 2019). This offset can result from both "horizontal" and "vertical" features (Figure 1), and can exceed several degrees centigrade in annual averages. For example, Kearney (2019) modeled coarse-scale soil temperatures at various depths considering the vertical features affecting the radiation balance. These vertical features include the effects of vegetation characteristics (e.g. structure and cover), snow cover and soil characteristics (e.g. moisture content, geological types, texture and bulk density; Lembrechts, Lenoir et al., 2019; Li, 1926; Zhang, Wang, Barr, & Black, 2008). The result of these vertical features is not only an instantaneous temperature offset between air and soil temperatures but also a buffering effect, that is, the temporal variability in temperature changes is lower in the soil than in the air (Ashcroft & Gollan, 2013; Geiger, 1950). Horizontal processes, on the other hand, relate more to the spatial resolution of the climatic data. They can be broken up into those that require only fine-resolution environmental information for specific sites (e.g. effects of slope and aspect on radiation balances; Bennie, Huntley, Wiltshire, Hill, & Baxter, 2008), and those where temperatures are also affected by neighboring locations (e.g. topographic shading, cold-air drainage and atmospheric temperature inversions, which are landscape context dependent; Ashcroft & Gollan, 2012; Whiteman, 1982).

How horizontal and vertical features interact to define differences between soil and air temperature may differ with the biome, season and day time. For example, in grasslands during summer, incoming short-wave solar radiation is usually the dominant factor determining daytime soil surface temperatures, which, in turn, result in higher air temperatures through convective heating (Geiger, 1950). However, during winter, horizontal processes such as cold-air drainage and coastal buffering can have larger effects, especially on overnight air temperatures, when air temperatures may be driving soil temperatures rather than vice-versa (Vitasse, Klein, Kirchner, & Rebetez, 2017). In dense forests, the situation is even more complex: upper canopies block the bulk of short-wave solar radiation such that sub-canopy temperatures are determined by convective heat transfer between the air surrounding the canopy and direct conductance through physical contact of different parts of the canopy layer, in addition to the limited radiation that does permeate the canopy (Körner & Paulsen, 2004; Lenoir et al., 2017; Zellweger, De Frenne, Lenoir, Rocchini, & Coomes, 2019). As a result, horizontal processes such as passing fronts, and winds blowing in hotter or colder air from outside the forest, will in large part define the—dampened—temperature patterns under forest canopies (Ashcroft, Chisholm, & French, 2008).

3 | THE NEED FOR MICROCLIMATE DATA ACROSS THE FIELD OF ECOLOGY

Many organisms living in the soil and close to the soil surface (e.g. soil micro-organisms like fungi, ground arthropods, herbs, mosses, tree seedlings and small vertebrates) only experience fine-scale soil and/or near-surface temperatures, and thus likely relate less strongly to free-air temperatures (Lembrechts, Lenoir et al., 2019; Niittynen & Luoto, 2017; Randin, Vuissoz, Liston, Vittoz, & Guisan, 2009). This may be reflected in a species' distribution but also their morphology, physiology and behavior (de Boeck, Velde, Groote, & Nijs, 2016; Kearney, Shine, & Porter, 2009; Körner & Paulsen, 2004; Opedal, Armbruster, & Graae, 2015). Many species indeed survive, live and reproduce where average background climate appears unsuitable, and equally may be gone from sites within apparently suitable areas where microclimatic extremes exceed their limits (Suggitt et al., 2011). Without microclimate data, we not only lack information on the potential thermal heterogeneity that is available for species to thermoregulate in situ but also on the true magnitude of climate change that species will be exposed to (Maclean, Suggitt, Wilson, Duffy, & Bennie, 2017; Pincebourde, Murdock, Vickers, & Sears, 2016). Accurately predicting how species' ranges will shift under climate change requires a good understanding of the variety of climate niches truly available to them (Lenoir et al., 2017; Maclean, Hopkins, Bennie, Lawson, & Wilson, 2015). The latter requires both a good understanding of what defines current microclimates and of how climate change will interact with the drivers of microclimatic conditions (Maclean, 2019). Additionally, it is the soil temperature rather than the air temperature that defines many ecosystem functions in and close to the soil, like evapotranspiration, decomposition, root growth, biogeochemical cycling and soil respiration (Gottschall et al., 2019; Hursh et al., 2017; Medinets, Gasche, Kiese, Rennenberg, & Butterbach-Bahl, 2019; Pleim & Gilliam, 2009; Portillo-Estrada et al., 2016). Given the repeatedly proven sensitivity of many of these processes to temperatures (Coûteaux, Bottner, & Berg, 1995; Rosenberg, Kimball, Martin, & Cooper, 1990; Schimel et al., 1996), here again having accurate measurements will be of utmost importance. The carbon balance in boreal forests, for example, is largely dependent on soil thaw and thus soil rather than air temperatures (Goulden et al., 1998).

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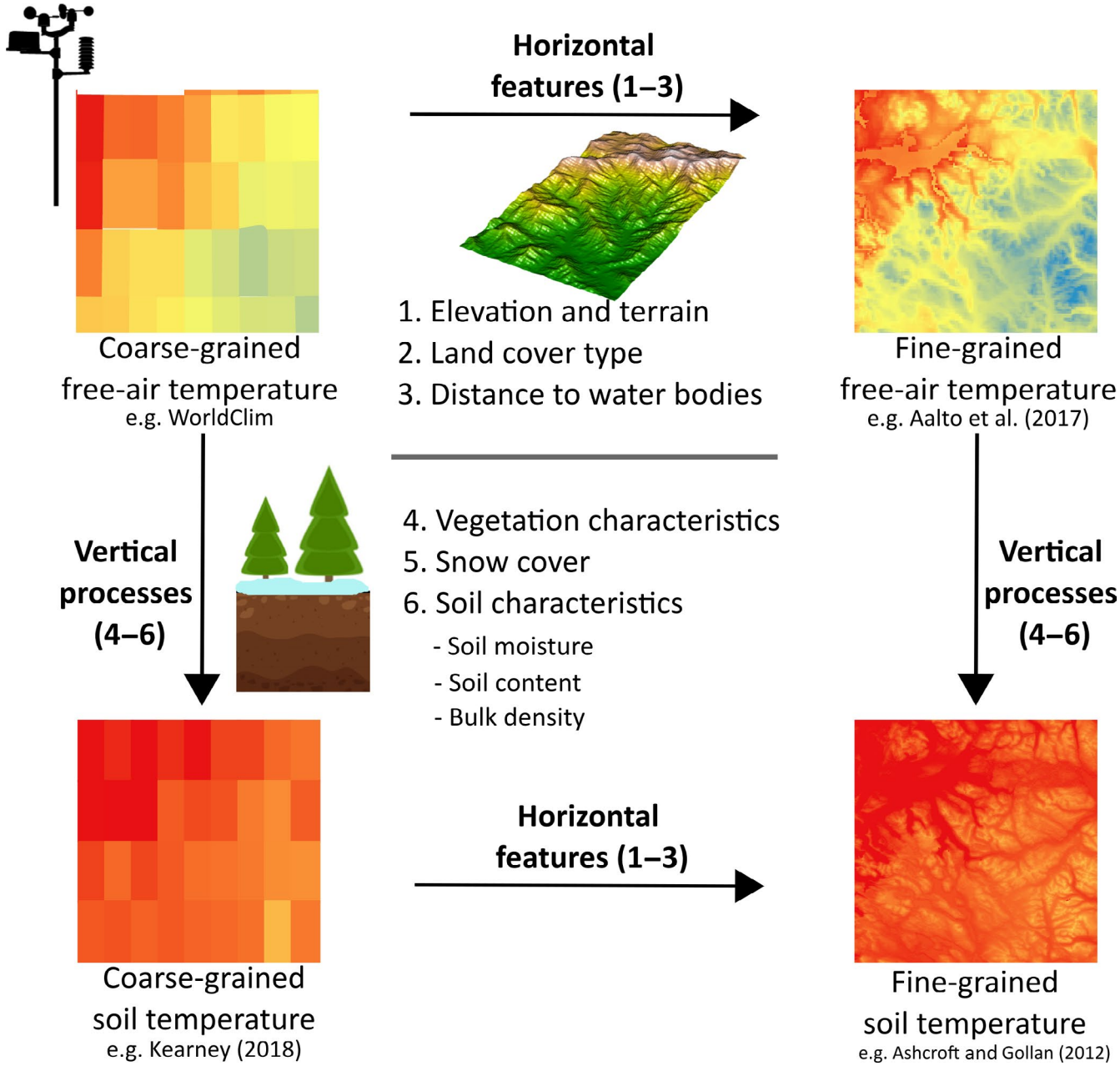

FIGURE 1 The horizontal and vertical drivers of the offset between in-situ soil and free-air temperatures. Conceptually, there are two different sets of features responsible for the offset between coarse-scale free air temperatures (top left, e.g. WorldClim, Fick & Hijmans, 2017) and fine-scale soil temperatures (bottom right, e.g. Ashcroft & Gollan, 2012; Lembrechts, Lenoir et al., 2019). First, one can incorporate fine-scale horizontal climate-forcing factors such as topography and terrain-related features, land cover types and distance to water bodies to go from coarse-scaled to finer resolutions (top right, e.g. Aalto, Riihimäki, Meineri, Hylander, & Luoto, 2017; Macek, Kopecký, & Wild, 2019). Second, one can consider observation height, and the effects of vegetation characteristics (like structure and cover), snow cover and soil characteristics (like moisture, geological types, texture and bulk density) on the radiation balance to convert from free-air to soil temperatures (e.g. Kearney, 2019). Both horizontal and vertical features can introduce positive or negative differences (offset values) between soil and air temperatures through their effects on processes related to the radiation balance, like wind, convective heat transfer and surface albedo. The complexities of these horizontal and vertical processes can vary with biome, season and time of day. Temperatures are represented here using an unspecified temperature range from cold (blue) to warm (red)

These realizations highlight the urgency to start using soil and near-surface microclimate data when modeling the ecology and biogeography of surface and soil-dwelling organisms, as well as the functioning of soil ecosystems, instead of readily available coarse-scaled free-air climate data (from e.g. CHELSA [Karger et al., 2017], TerraClimate [Abatzoglou, Dobrowski, Parks, & Hegewisch, 2018] or WorldClim [Fick & Hijmans, 2017]). While a suit of models now exist that produce fine-scale climate data (Bramer et al., 2018; Lembrechts, Nijs et al., 2019), we do not yet fully understand whether models using data that represent average conditions over

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large areas provide adequate "mean field approximations" of (i.e. are representative for) more complex spatiotemporal effects driven by the climatic conditions that organisms experience (Bennie, Wilson, Maclean, & Suggitt, 2014). To accomplish the latter, global in-situ data are needed for large-scale fine-resolution calibration and validation of these models. However, while the quality and resolution of free-air temperature data and models at the global scale are rapidly improving (Bramer et al., 2018), soil temperature datasets used in biogeography and biogeochemistry are still largely restricted to the landscape or regional scale, at best, and from intensively studied regions only (Aalto, Scherrer, Lenoir, Guisan, & Luoto, 2018; Ashcroft et al., 2008; Ashcroft, Chisholm, & French, 2009; Carter et al., 2015), or they are derived from models lacking fine-grained ground-truthing data (e.g. Copernicus Climate Change Service (C3s), 2019). Land surface temperatures as obtained from satellite data, on the other hand, are hampered by their inability to measure below the vegetation cover (Bramer et al., 2018).

To accurately describe and predict the (future) distribution and/or traits of surface and soil-dwelling species at larger scales, we need to improve our general knowledge of the offsets and spatiotemporal changes in variability between soil-level and free-air temperatures (Aalto et al., 2018; Lembrechts, Lenoir et al., 2019). There is an urgent need to work toward globally available soil and near-surface temperature data based on in-situ measurements and at relevant spatiotemporal resolutions (Ashcroft & Gollan, 2012; Meineri & Hylander, 2017; Opedal et al., 2015; Pradervand, Dubuis, Pellissier, Guisan, & Randin, 2014; Slavich, Warton, Ashcroft, Gollan, & Ramp, 2014).

4 | LAUNCH OF THE SoilTemp DATABASE

To tackle these issues, we launch an ambitious database initiative, compiling soil and near-surface temperature data from all over the world into a global geospatial database: SoilTemp. At the time of writing, we brought together temperature data from 7,538 sensors placed both below, at and above (up to 2 m) the soil surface (Figure 2a), which is an accumulation of over 180,000 months of temperature data with measurement intervals between 1 and 240 min (>30% every 60 min). The database hosts loggers from 51 different countries spread across all continents, with a broad distribution across the world's climatic space (Figure 2b). There is a dominance of time series from Europe and areas below 1,500 m a.s.l. (Figure 2c,d). More than 75% of sensor measurements occurred within the last decade, but the database does contain several time series covering longer time periods as well, with a maximum of 42 years (Figure 2d).

When the remaining critical gaps in our spatial coverage will be filled (see below), this database will allow global assessments of the long-established theories on boundary layer climatology in heterogeneous environments (Geiger, 1950), which has so far been lacking. The growing database provides a unique opportunity to disentangle the role of the different horizontal and vertical features influencing soil and near-surface temperature across all biomes of the world, with high spatial and temporal resolutions. It will allow relating patterns in soil temperature to processes in the lower air layers and calibrate and validate global models of soil temperature and (micro) climate (Carter et al., 2015; Kearney, Isaac, & Porter, 2014; Kearney, Shamakhy, et al., 2014; Maclean et al., 2017). It will also allow us to create global maps of a wide array of general and microclimate-specific bioclimatic variables (e.g. growing degree days, growing season length) at relevant spatiotemporal resolutions (Körner & Hiltbrunner, 2018).

Ultimately, this joint global effort and the resulting global microclimatic products will enable us to improve analyses of the relationships between species' macroecology and the microclimate they experience, identify microrefugia and stepping stones and improve global models of ecosystem functioning and element cycling. Indeed, replacing the coarse-scaled free-air temperature averages used traditionally in models in all fields of ecology with these more relevant soil-specific data products is likely to increase their descriptive and predictive power, as the countless above-mentioned regional studies exemplify (Lembrechts, Lenoir et al., 2019). Additionally, this first global effort to combine and collect in-situ measurements will help solve long-standing issues regarding sensor comparability and data collection variability (Bramer et al., 2018), as well as address the question at what spatial scale microclimate data can prove most informative for ecological modeling (Jucker et al., 2020). The temperature time series in the database, many of which are covering increasingly long time periods of up to a decade or more, will also allow fine-tuning forecasts of microclimate data into the future by deepening our understanding of the link between microclimatic dynamics in the soil and the air (Bramer et al., 2018; Lenoir et al., 2017; Maclean, 2019; Wason, Bevilacqua, & Dovciak, 2017), improving our predictions of biodiversity and ecosystem functioning under climate change.

5 | DIG OUT YOUR LOGGERS! A CALL FOR CONTRIBUTIONS

To reach these goals, we encourage scientists owning in-situ measured temperature data to submit these to the growing SoilTemp database. All time series spanning 1 month or more, with temperature measurements a maximum of 4 hr apart, all soil depths, all heights above the ground up till 2 m, all biomes, and all sensor types and brands will be accepted. Note that both spatially dense and sparse logger networks, as well as single loggers are accepted. The achieved spatial resolution is dependent on the provision of spatially precise coordinates to achieve a good relationship with potential explanatory variables (e.g. high-resolution remotely sensed environmental data). If we have these coordinates and thus the location and distance between loggers, we can effectively obtain the extent and spacing for each logger network (Western, Grayson, & Blöschl, 2002).

We include data from both observational and experimental plots, yet sensors have to be measuring in-situ and not in pots, and experiments manipulating the local climate (e.g. open-top chambers, rain-out shelters or vegetation-removal experiments) are excluded

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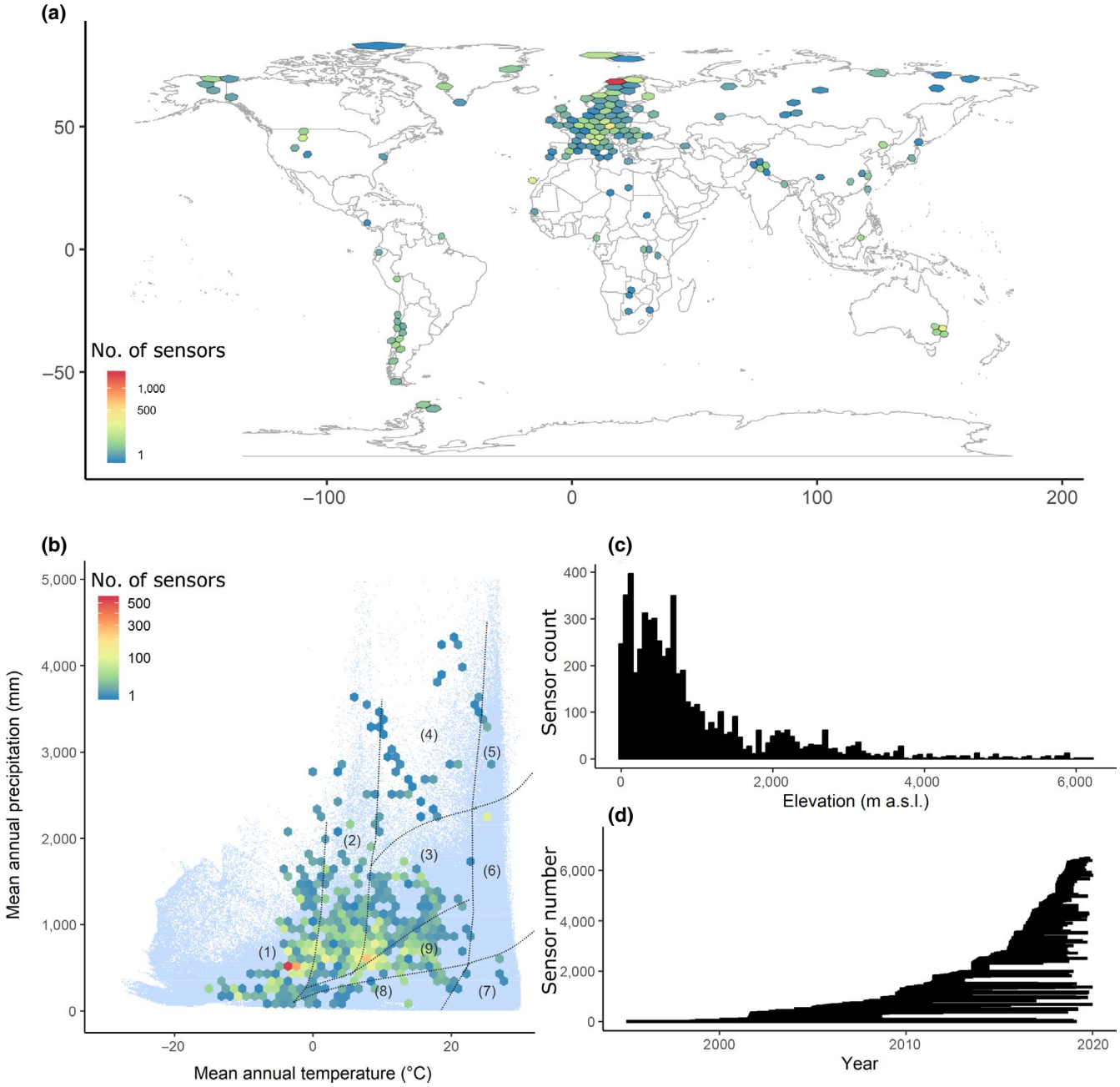

FIGURE 2 Overview of the status of the SoilTemp-database as of March 2020. Spatial (a), climatic (b), elevational (c) and temporal (d) distribution of sensors in the SoilTemp-database as of March 2020. (a) Background world map in WGS1984, hexagons with a resolution of approximately 70,000 km^2 using the dggridR-package in R. (b) Colors of hexagons indicate the number of sensors at each climatic location, with a 40 × 40 bin resolution. Small dots in the background represent the global variation in climatic space (obtained by sampling 1,000,000 random locations from the CHELSA world maps at a spatial resolution of 2.5 arc minutes. Overlay with dotted lines and numbers (from 1 to 9) depict a delineation of Whittaker biomes (adapted from Whittaker, 1975): (1) tundra and ice, (2) boreal forest, (3) temperate seasonal forest, (4) temperate rainforest, (5) tropical rainforest, (6) tropical seasonal forest/savanna, (7) subtropical desert, (8) temperate grassland/desert, (9) woodland/shrubland. (c) Number of sensors in each elevation class. (d) Time span covered by each sensor in the database, ranked by starting date. Data showed from 1992 onwards, note that the time period covered by four loggers with starting dates in 1976 is truncated

(Table 1). Given currently less well-represented climate regions, we especially encourage submissions from extreme cold and hot environments to fill the remaining gaps in our global coverage. More specifically, hot tropical climates (both tropical rainforests and tropical seasonal forests and savannas) and cold and hot deserts are currently still largely underrepresented (Figure 2b), in particular from Africa, Asia, Antarctica and the Americas (Figure 2a). Data contributors will be invited as co-authors on the main global papers resulting from this database (see Supporting Information for details on terms of use and data ownership).

By encouraging sampling and submissions from remote areas, we aim to help solve the global sampling bias in soil ecological

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TABLE 1 Minimal data requirements and obligatory metadata for submission to the database. For more details, see Supporting Information

Minimum data requirements	Obligatory metadata
Minimum one consecutive month of in-situ measured temperature time series	Accurate (handheld GPS or finer) spatial coordinates of the loggers (+ estimated accuracy)
Maximum time interval between measurements: 4 hr	Height/depth of the sensor relative to the soil surface
No climate manipulation experiments (only control plots of those experiments or observational studies)	Type or brand of temperature sensor used, and type of shelter (e.g. no shelter, home-made shelter, Stevenson screen, etc.)
No modeling studies (only empirical data)	Temporal resolution of the sensor
	Habitat classification

data (Cameron et al., 2018; Guerra et al., 2019), and we hope to build a truly global network representing—and actively engaging—scientists from a wide diversity of cultural backgrounds (Maestre & Eisenhauer, 2019). More information is available on the SoilTemp website, accessible via Figshare (https://doi.org/10.6084/m9.figshare.12126516).

When fully established, the SoilTemp database and its derivative products (e.g. bioclimatic variables) will be made freely available to facilitate the analysis of global patterns in microclimates, increase the comparability between regional studies and simplify the use of accurate microclimatic data in ecology (Bramer et al., 2018). At the moment, critical metadata are already freely accessible via Figshare (https://doi.org/10.6084/m9.figshare.12126516). Given the absence of and the need for globally available soil microclimate data products at relevant spatial resolutions for use in ecological analyses, we believe that SoilTemp has the potential to become a highly important resource that will enable a step change in ecological modeling.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

AUTHOR CONTRIBUTION

J.J.L. performed the analyses and wrote the manuscript, J.J.L., J.A., M.B.A., P.D.F., M.K., J.L., M.L., I.M.D.M. and I.N. led the consortium and contributed to the writing; all authors contributed to the consortium and provided editorial advice.

ORCID

Jonas J. Lembrechts https://orcid.org/0000-0002-1933-0750
Juha Aalto https://orcid.org/0000-0001-6819-4911
Michael B. Ashcroft https://orcid.org/0000-0003-2157-5965
Pieter De Frenne https://orcid.org/0000-0002-8613-0943
Martin Kopecký https://orcid.org/0000-0002-1018-9316
Jonathan Lenoir https://orcid.org/0000-0003-0638-9582
Ilya M. D. Maclean https://orcid.org/0000-0001-8030-9136
Olivier Roupsard https://orcid.org/0000-0002-1319-142X
Juha M. Alatalo https://orcid.org/0000-0001-5084-850X
Stuart W. Smith https://orcid.org/0000-0001-9396-6610
Robert G. Björk https://orcid.org/0000-0001-7346-666X
Lena Muffler https://orcid.org/0000-0001-8227-7297
Felix Gottschall https://orcid.org/0000-0002-1247-8728
Josef Urban https://orcid.org/0000-0003-1730-947X
Martin Svátek https://orcid.org/0000-0003-2328-4627
Shyam S. Phartyal https://orcid.org/0000-0003-3266-6619
Sonja Wipf https://orcid.org/0000-0002-3492-1399
Nico Eisenhauer https://orcid.org/0000-0002-0371-6720
Mihai Pușcaș https://orcid.org/0000-0002-2632-640X
Pavel D. Turtureanu https://orcid.org/0000-0002-7422-3106
Andrej Varlagin https://orcid.org/0000-0002-2549-5236
Alistair S. Jump https://orcid.org/0000-0002-2167-6451
Krystal Randall https://orcid.org/0000-0003-2507-1000
Ellen Dorrepaal https://orcid.org/0000-0002-0523-2471
Josefine Walz https://orcid.org/0000-0002-0715-8738
Miroslav Svoboda https://orcid.org/0000-0003-4050-3422
Salvatore R. Curasi https://orcid.org/0000-0002-4534-3344
Austin Koontz https://orcid.org/0000-0002-6103-5894

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William D. Pearse https://orcid.org/0000-0002-6241-3164
Bente Jessen Graae https://orcid.org/0000-0002-5568-4759
Toke T. Høye https://orcid.org/0000-0001-5387-3284
Juan Lorite https://orcid.org/0000-0003-4617-8069
Michele Carbognani https://orcid.org/0000-0001-7701-9859
Marcello Tomaselli https://orcid.org/0000-0003-4208-3433
T'ai G. W. Forte https://orcid.org/0000-0002-8685-5872
Stef Haesen https://orcid.org/0000-0002-4491-4213
Ben Somers https://orcid.org/0000-0002-7875-107X
Koenraad Van Meerbeek https://orcid.org/0000-0002-9260-3815
Mana Gharun https://orcid.org/0000-0003-0337-7367
Nina Buchmann https://orcid.org/0000-0003-0826-2980
Jiri Dolezal https://orcid.org/0000-0002-5829-4051
Andrew D. Thomas https://orcid.org/0000-0002-1360-1687
Dany Ghosn https://orcid.org/0000-0003-1898-9681
Miguel A. de Pablo https://orcid.org/0000-0002-4496-2741
Julia Kemppinen https://orcid.org/0000-0001-7521-7229
Pekka Niittynen https://orcid.org/0000-0002-7290-029X
Lisa Rew https://orcid.org/0000-0002-2818-3991
Christian Larson https://orcid.org/0000-0002-7567-4953
James D. M. Speed https://orcid.org/0000-0002-0633-5595
Jonas Ardö https://orcid.org/0000-0002-9318-0973
Maaike Y. Bader https://orcid.org/0000-0003-4300-7598
Rafaella Canessa https://orcid.org/0000-0002-6979-9880
Angela Stanisci https://orcid.org/0000-0002-5302-0932
Juergen Kreyling https://orcid.org/0000-0001-8489-7289
Valeria Aschero https://orcid.org/0000-0003-3865-4133
Marek Čiliak https://orcid.org/0000-0002-6720-9365
František Máliš https://orcid.org/0000-0003-2760-6988
Pallieter De Smedt https://orcid.org/0000-0002-3073-6751
Sanne Govaert https://orcid.org/0000-0002-8939-1305
Camille Meeussen https://orcid.org/0000-0002-5869-4936
Pieter Vangansbeke https://orcid.org/0000-0002-6356-2858
Andrea Lamprecht https://orcid.org/0000-0002-8719-026X
Harald Pauli https://orcid.org/0000-0002-9842-9934
Klaus Steinbauer https://orcid.org/0000-0002-3730-9920
Manuela Winkler https://orcid.org/0000-0002-8655-9555
Masahito Ueyama https://orcid.org/0000-0002-4000-4888
Tudor-Mihai Ursu https://orcid.org/0000-0002-4898-6345
Sylvia Haider https://orcid.org/0000-0002-2966-0534
Ronja E. M. Wedegärtner https://orcid.org/0000-0003-4633-755X
Martin Wilmking https://orcid.org/0000-0003-4964-2402
Jan Altman https://orcid.org/0000-0003-4879-5773
Josef Brůna https://orcid.org/0000-0002-4839-4593
Martin Macek https://orcid.org/0000-0002-5609-5921
Matěj Man https://orcid.org/0000-0002-4557-8768
Jan Wild https://orcid.org/0000-0003-3007-4070
Pascal Vittoz https://orcid.org/0000-0003-4218-4517
Peter Barančok https://orcid.org/0000-0003-1171-2524
Róbert Kanka https://orcid.org/0000-0002-7071-7280
Andrej Palaj https://orcid.org/0000-0001-7054-4183
Marijn Bauters https://orcid.org/0000-0003-0978-6639
José-Luis Benito Alonso https://orcid.org/0000-0003-1086-8834
Valter Di Cecco https://orcid.org/0000-0001-9862-1267
Zuzana Sitková https://orcid.org/0000-0001-6354-6105
Liesbeth van den Brink https://orcid.org/0000-0003-0313-8147
Robert Weigel https://orcid.org/0000-0001-9685-6783
C. Johan Dahlberg https://orcid.org/0000-0003-0271-3306
Sergiy Medinets https://orcid.org/0000-0001-5980-1054
Volodymyr Medinets https://orcid.org/0000-0001-7543-7504
Miguel Portillo-Estrada https://orcid.org/0000-0002-0348-7446
Lore T. Verryckt https://orcid.org/0000-0002-9452-5216
Ann Milbau https://orcid.org/0000-0003-3555-8883
Gergana N. Daskalova https://orcid.org/0000-0002-5674-5322
Haydn J. D. Thomas https://orcid.org/0000-0001-9099-6304
Isla H. Myers-Smith https://orcid.org/0000-0002-8417-6112
Jörg G. Stephan https://orcid.org/0000-0001-6195-7867
Patrice Descombes https://orcid.org/0000-0002-3760-9907
Florian Zellweger https://orcid.org/0000-0003-1265-9147
Esther R. Frei https://orcid.org/0000-0003-1910-7900
Bernard Heinesch https://orcid.org/0000-0001-7594-6341
Christopher Andrews https://orcid.org/0000-0003-2428-272X
Rebecca A. Senior https://orcid.org/0000-0002-8208-736X
Juan J. Jimenez https://orcid.org/0000-0003-2398-0796
Julia Boike https://orcid.org/0000-0002-5875-2112
Thomas Scholten https://orcid.org/0000-0002-4875-2602
Brett Scheffers https://orcid.org/0000-0003-2423-3821
Edmund W. Basham https://orcid.org/0000-0002-0167-7908
Jian Zhang https://orcid.org/0000-0003-0589-6267
Charly Géron https://orcid.org/0000-0001-7912-4708
Fatih Fazlioglu https://orcid.org/0000-0002-4723-3640
Onur Candan https://orcid.org/0000-0002-9254-4122
Jhonatan Sallo Bravo https://orcid.org/0000-0001-9007-4959
Filip Hrbacek https://orcid.org/0000-0001-5032-9216
Kamil Laska https://orcid.org/0000-0002-5199-9737
Edoardo Cremonese https://orcid.org/0000-0002-6708-8532
Peter Haase https://orcid.org/0000-0002-9340-0438
Fernando E. Moyano https://orcid.org/0000-0002-4090-5838
Christian Rossi https://orcid.org/0000-0001-9983-8898
Ivan Nijs https://orcid.org/0000-0003-3111-680X

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